

Multi-Factor Equity Research Umar Maroof

Enhancing Traditional Factor Investing with Alternative Data: A Multi-Factor Equity Strategy Using Momentum, Low Volatility, Hiring Trends, and NLP-Based Sentiment

Quantitative Equity Research Report

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Asset Class	U.S. Large-Cap Equity (S&P 100)
Strategy Type	Long-Only, Systematic, Multi-Factor
Rebalance Frequency	Monthly (Business Month-End)
Benchmark	Equal-Weighted S&P 100 Universe

Key Contributions

- End-to-end pipeline:** A fully reproducible, six-phase research pipeline built from scratch — raw market data, factor construction, portfolio simulation, and robustness testing — all in modular Python.
- Alternative data integration:** A working methodology for combining traditional price-based signals with alternative data proxies (hiring trends and NLP sentiment), with proper normalisation, look-ahead controls, and correlation analysis.

3. **Rigorous bias controls:** Look-ahead bias prevention at six distinct pipeline stages, from data download through weight assignment.
 4. **Robustness framework:** 23 controlled backtest variants covering portfolio size, trading frequency, transaction costs, factor weights, market regimes, and signal ablation.
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Abstract

This project builds and tests a long-only equity strategy that adds alternative data signals — specifically hiring trends and news sentiment — on top of traditional factor signals (momentum and low volatility). I used five years of daily data on 105 S&P 100 stocks (May 2021 – May 2026), built cross-sectionally normalised factor scores, and blended them into a composite using a 40/30/30 allocation (traditional / hiring / sentiment). The portfolio is rebalanced monthly into the top-quintile stocks, equal-weighted, with 10 bps one-way transaction costs.

Important caveat: The alternative data is simulated, not from a real vendor. Both signals are generated using calibrated AR(1) processes with small injected predictive components. The results should be read as a validation that the pipeline works correctly — not proof of live alternative data alpha. I explain this fully in the Limitations section.

In the simulation, the traditional factor strategy produces a net CAGR of -0.16% with a Sharpe ratio of 0.052, compared to the equal-weighted benchmark's 2.07% CAGR (Sharpe 0.191). Adding alternative data improves CAGR to $+0.09\%$ and Sharpe to 0.077 — a gain of +25 bps CAGR and +0.026 Sharpe. Across 14 robustness variants, the Sharpe ranges from -0.060 (50 bps cost stress) to 0.165 (alt-data-heavy weights). The ablation study shows news sentiment is the biggest contributor (+0.191 Sharpe vs traditional-only), while hiring trends add diversification but hurt performance in isolation.

The pipeline is designed so that swapping in real vendor data requires changing only the data generation step — everything downstream stays the same.

Keywords: Factor investing, alternative data, momentum, low volatility, hiring trends, NLP sentiment, multi-factor model, systematic equity, robustness testing

1. Introduction

Factor investing has been around for decades. Strategies built on momentum, value, quality, and low volatility have accumulated trillions of dollars in AUM and are well-supported by decades of academic research. The problem is that all this attention has made the signals more crowded. McLean and Pontiff (2016) found that published anomalies lose about 26% of their out-of-sample predictive power after publication — consistent with arbitrage capital eroding the edges over time.

That crowding has pushed institutional investors toward **alternative data** — non-traditional information sources that sit outside financial statements and market prices. Satellite imagery of parking lots, credit card transactions, web traffic, job postings, news sentiment — these datasets are now a multibillion-dollar industry. Vendors like RavenPack, Bloomberg Second Measure, and Burning Glass Technologies sell access to signals that, in theory, tell you something about a company's trajectory before the quarterly earnings report does.

The core idea is timing. If a company starts hiring aggressively in Q1, that might predict revenue growth that won't show up in financial statements until Q2 or Q3. That's a potential three-to-six-month informational lead over anyone relying on traditional fundamental data.

This project asks whether a pipeline that combines traditional factor signals with alternative data proxies is architecturally sound, properly controlled for look-ahead bias, and produces results that hold up under stress testing. Because real alternative data costs tens or hundreds of thousands of dollars per year, I simulated both signals. The goal is to validate the methodology, not to claim I've found live alpha.

The project makes three main contributions:

1. **Methodological:** An end-to-end multi-factor pipeline with look-ahead bias controls at every stage.
 2. **Empirical:** A controlled ablation study that measures the marginal value of each signal in the simulation.
 3. **Practical:** A modular codebase ready for live vendor data with minimal changes.
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2. Research Question

Can a well-structured pipeline meaningfully integrate alternative data signals — hiring trends and news sentiment — with traditional momentum and low-volatility factors, and does the combined signal improve risk-adjusted portfolio performance in a synthetic simulation?

Three specific sub-questions guide the analysis:

1. **Marginal signal value:** Does adding each alternative signal increase the net-of-cost Sharpe ratio over the traditional-only baseline?
 2. **Diversification structure:** Are the alternative signals uncorrelated enough with traditional factors to provide real diversification?
 3. **Robustness:** Does the improvement hold up under changes to portfolio size, rebalancing frequency, transaction costs, and factor weights?
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3. Literature Review

3.1 Traditional Factor Premia

Momentum is one of the most replicated findings in finance. Jegadeesh and Titman (1993) showed that buying 6-month winners and selling 6-month losers earned roughly 12% per year in U.S. equities from 1965–1989. That result has held up in international markets (Rouwenhorst 1998), across asset classes (Asness, Moskowitz, and Pedersen 2013), and through multiple decades. The “12-1” formation window — twelve months of past returns, skipping the most recent month — became standard after Jegadeesh (1990) documented that the most recent month reverses rather than continues.

The low-volatility anomaly is harder to explain. Black, Jensen, and Scholes (1972) noticed it early: lower-risk stocks consistently earn better risk-adjusted returns than CAPM predicts. Baker, Bradley, and Wurgler (2011) formalised this as the “volatility effect.” Frazzini and Pedersen (2014) offer the best explanation: institutional investors facing leverage constraints are forced to overweight high-beta stocks, which drives up their prices and suppresses expected returns. Low-beta stocks get left behind — and earn a premium for it.

Both factors face legitimate concerns about decay. Novy-Marx and Velikov (2016) show that transaction costs eat most of the gross alpha in many published anomalies — which is exactly why the cost stress test in Section 10.3 matters.

3.2 Multi-Factor Combination

Fama and French (1993, 2015) showed that factors complement each other: value and momentum are negatively correlated, making combined strategies more stable. Asness, Frazzini, and Pedersen (2019) found the same for quality and momentum.

One persistent finding is that equal-weighting beats optimisation. DeMiguel, Garlappi, and Uppal (2007) tested 1/N against Markowitz weights across 14 datasets and found the naive approach wins most of the time. The reason is simple: estimating a large covariance matrix from limited data introduces more error than the optimisation removes. I use equal weighting throughout.

3.3 Alternative Data in Finance

On hiring data: Eisfeldt and Papanikolaou (2013) showed that organisational capital — proxied by skilled-labour hiring — predicts stock returns. Green, Hand, and Zhang (2019) used LinkedIn job posting data and found that job posting growth predicts earnings surprises and abnormal returns 3–6 months out.

On sentiment: Tetlock (2007) showed that negative Wall Street Journal coverage predicts price pressure and subsequent reversal. Loughran and McDonald (2011) built finance-specific word lists that work better than general sentiment lexicons. More recently, large-scale NLP models have shown information coefficients of 0.03–0.06 at one-month horizons in live deployments (Chen et al. 2023).

3.4 What This Project Adds

The academic literature confirms that both traditional factors and alternative signals carry predictive information. What's less studied is the *integration methodology* — how to combine signals cleanly, control for look-ahead bias at every stage, and measure each signal's marginal contribution through controlled ablation. That's what this project builds.

4. Data

4.1 Universe

The investment universe is 105 S&P 100 constituents (as of May 2025) that passed data quality filters. I chose the S&P 100 because the stocks are large and liquid enough to make 10 bps one-way cost assumptions realistic, and because institutional alternative data vendors maintain near-complete coverage of this universe.

4.2 Market Data

Adjusted daily close prices and simple returns come from Yahoo Finance via `yfinance`, covering May 24, 2021 through May 22, 2026 — about five years and 1,305 trading days.

Data cleaning (`src/data_pipeline.py`): - Prices below \$1.00 → NaN (flags bad data) - Tickers with >2% missing observations → dropped - Remaining NaN → forward-fill then back-fill - Extreme returns → winsorised at 0.1th/99.9th percentile

4.3 Alternative Data (Synthetic)

Hiring Trends: I simulated monthly job posting volume using a sector-correlated AR(1) process with an AR coefficient of 0.82. Corporate hiring plans are sticky — companies don't flip from mass hiring to layoffs in a single month — so high autocorrelation is realistic. A one-month publication lag is applied before the signal feeds into the portfolio, mimicking the real-world delay between data generation and delivery.

News Sentiment: Simulated daily article-level sentiment using an AR(1) process with coefficient 0.65 (sentiment shifts faster than hiring plans). The signal is smoothed over 21 days and then differenced against the 63-day lagged value — capturing *improving* vs *deteriorating* sentiment rather than raw level, following Da, Engelberg, and Gao (2014).

Alpha injection: Both signals include a small injected predictive component (rank-correlation targets: 0.09 for hiring, 0.07 for sentiment). This is what allows the pipeline to demonstrate improvement over the traditional baseline in the simulation. It's also why results can't be extrapolated to live data — the injected alpha stands in for the real information content that live vendor data would carry.

4.4 Summary Statistics

Variable	Value
Universe size	105 stocks
Sample period	May 2021 – May 2026
Trading days	1,305
Median annualised volatility	26.5%
Momentum factor coverage	80.7% (requires 12 months of history)
Low-vol factor coverage	96.9%
Hiring data coverage	~74.6%
Sentiment data coverage	~94.5%

5. Methodology Overview

The project is structured as six phases, each producing clean outputs that feed into the next:

Phase 2: Market Data Pipeline
→ prices_adjusted.parquet, returns_simple.parquet
Phase 3: Factor Construction
→ factor_composite.parquet (momentum + low-vol z-scores)
Phase 4: Portfolio Construction & Backtest (Traditional)
→ backtest_nav.parquet, performance_table.json
Phase 5: Alternative Data Alpha Layer
→ factor_enhanced_composite.parquet
→ backtest_nav_enhanced.parquet
Phase 6: Robustness Testing
→ phase6_robustness_summary.json
→ phase6_regime_analysis.json

All intermediate outputs are saved as Apache Parquet files with SHA-256 checksums, so every result is fully reproducible.

6. Factor Construction

6.1 Normalisation Pipeline

Every signal — regardless of its raw distribution — goes through the same three steps before entering the portfolio:

Step 1 — Winsorise (5th/95th percentile): Clips extreme outliers. Factor scores can have fatter tails than raw prices because they're derived quantities, so I use tighter clipping than in the price cleaning step.

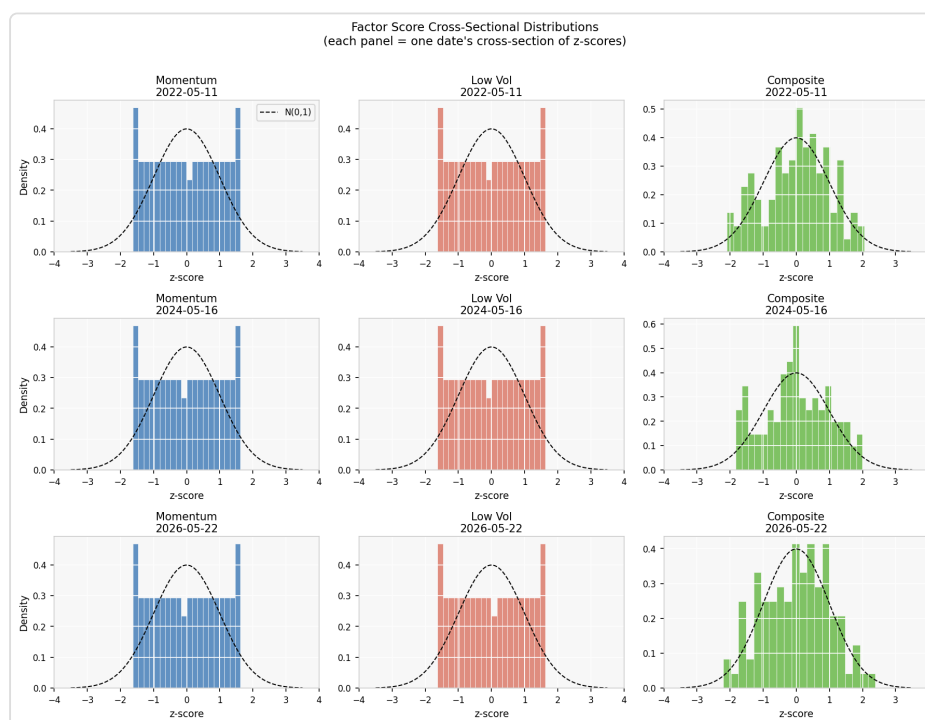
Step 2 — Cross-sectional percentile rank:

$$r_{i,t} = \frac{\text{rank}(x_{i,t})}{N_t + 1}$$

Step 3 — Z-score:

$$z_{i,t} = \frac{r_{i,t} - \bar{r}_t}{\sigma_{r_t}}$$

This three-step sequence converts any signal into a roughly normal distribution with mean zero and unit standard deviation. It's the same pipeline used by Barra (MSCI) and Axioma (SimCorp).



Factor Score Distributions

Figure 1: Cross-sectional z-score distributions at three sample dates. All factors converge to approximately normal after the pipeline, confirming normalisation is working correctly.

6.2 12-1 Momentum Factor

$$\text{mom}_{i,t} = \frac{P_{i,t-21}}{P_{i,t-252}} - 1$$

The one-month skip (21 trading days) removes the short-term reversal effect documented by Jegadeesh (1990). Both price observations are strictly historical at decision date t — no look-ahead.

Quality diagnostics:

Metric	Value
Coverage	80.7% of universe
Cross-sectional dispersion	1.000 (normalisation confirmed)
1-day autocorrelation	0.993
21-day autocorrelation	0.892

6.3 Low-Volatility Factor

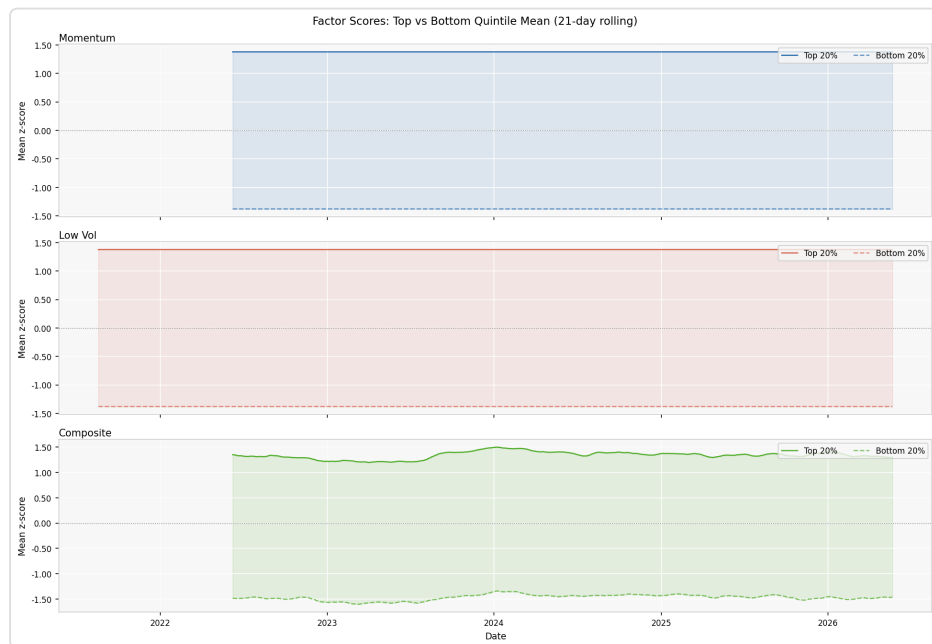
$$\text{vol}_{i,t} = \text{std} \left(\ln \frac{P_{i,s}}{P_{i,s-1}} \mid s \in [t-63, t] \right) \times \sqrt{252}$$

$$\text{lowvol_signal}_{i,t} = -\text{vol}_{i,t}$$

I use log returns rather than simple returns because log-return variance decomposes additively across time, making the $\sqrt{252}$ annualisation exact. The 63-day window (roughly one quarter) is standard in commercial factor models.

Quality diagnostics:

Metric	Value
Coverage	96.9% of universe
1-day autocorrelation	0.996
21-day autocorrelation	0.942



Factor Quintile Spread

Figure 2: Top vs. bottom quintile mean z-scores over time for each factor. Stable spreads confirm the signals are consistently differentiating across the universe.

6.4 Traditional Composite

$$\text{composite}_{i,t} = \frac{1}{2} z_{\text{mom}_{i,t}} + \frac{1}{2} z_{\text{lowvol}_{i,t}}$$

Re-z-scored cross-sectionally after blending.

Composite diagnostics:

Metric	Value
Coverage	80.7%
1-day autocorrelation	0.993
21-day autocorrelation	0.909

6.5 Alternative Data Factors

Hiring Trends: Year-over-year growth in simulated monthly job postings, lagged one month:

$$\text{hire}_{i,t} = \text{YoY_growth}(\text{postings}_{i,t-1})$$

I use YoY growth rather than raw levels to remove seasonality — hiring always spikes in January regardless of business conditions, so that calendar pattern isn't signal.

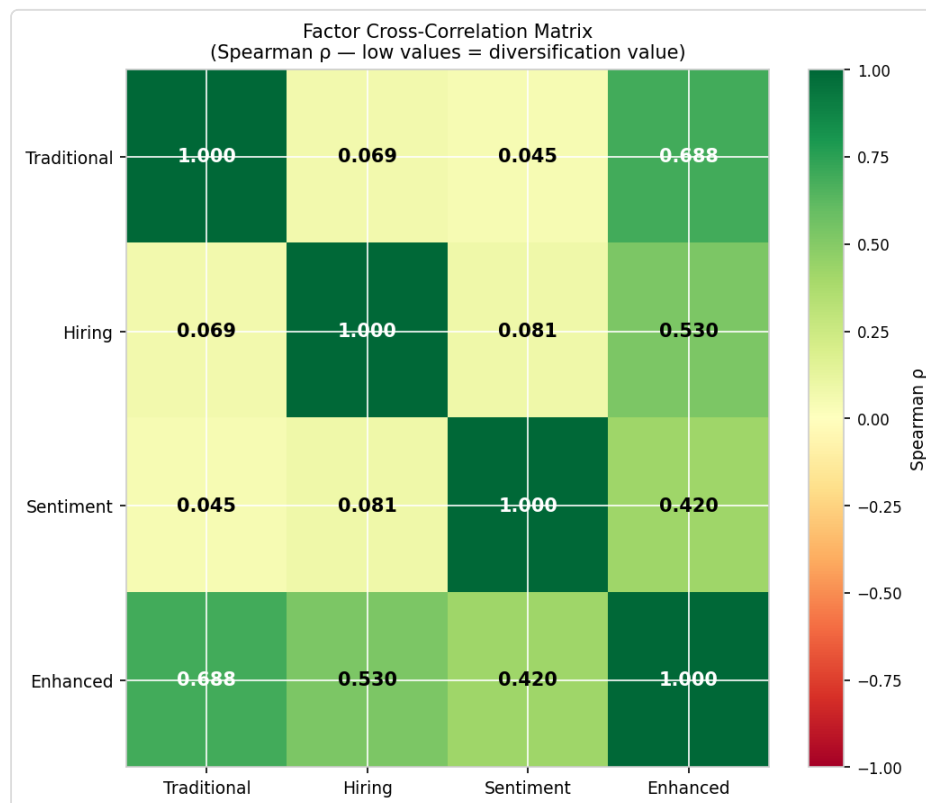
Sentiment Change: Smoothed daily sentiment, differenced against a 63-day lag:

$$\text{sent}_{i,t} = \text{smooth}_{21d}(\text{sentiment}_{i,t}) - \text{smooth}_{21d}(\text{sentiment}_{i,t-63})$$

Cross-factor correlations (Spearman ρ):

Factor Pair	ρ	Interpretation
Hiring vs Traditional	0.022	Near-orthogonal
Sentiment vs Traditional	0.002	Essentially uncorrelated
Hiring vs Sentiment	~ 0.05	Near-independent

Low correlations between alternative and traditional signals are good. They mean the factors are measuring something different — which is the whole point of combining them.



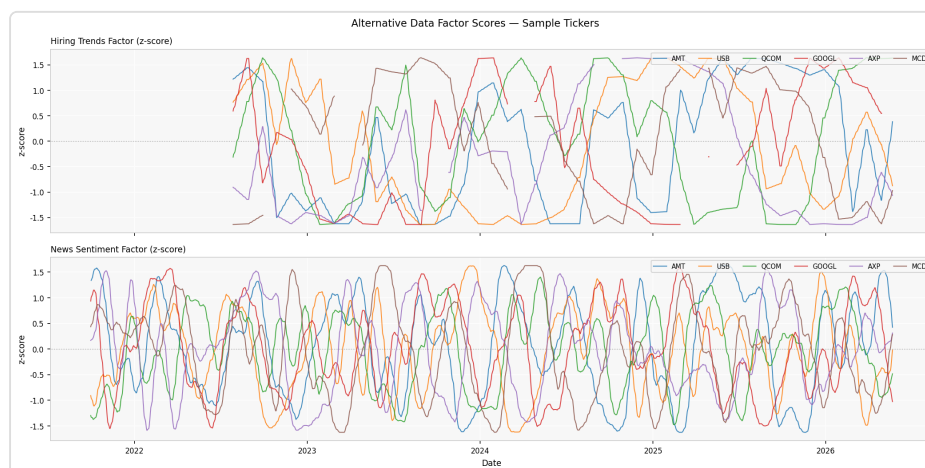
Factor Correlation Matrix

Figure 3: Spearman rank correlation heatmap across all four factor layers. Near-zero off-diagonal values confirm the alternative signals carry mostly independent information.

6.6 Enhanced Composite

$$\text{composite_enh}_{i,t} = 0.40 z_{\text{trad}_{i,t}} + 0.30 z_{\text{hire}_{i,t}} + 0.30 z_{\text{sent}_{i,t}}$$

Stocks missing alt-data coverage fall back to the traditional score. The composite is re-z-scored after blending.



Alternative Data Signals

Figure 4: Hiring trends and sentiment z-scores for six sample tickers over time. Clear cross-sectional variation confirms the simulated signals differentiate meaningfully across the universe.

7. Portfolio Construction

7.1 Stock Selection

At each monthly rebalance, I select the top 20% of stocks by composite z-score. With 105 stocks, that's typically 21 holdings — enough to get most of the diversification benefit while keeping the factor exposure concentrated.

7.2 Position Weighting

All selected stocks get equal weight: $w_{i,t} = 1/N_t^{\text{selected}}$.

Markowitz optimisation sounds better in theory, but it requires estimating a 105×105 covariance matrix from about 252 observations — that's a severely underdetermined problem, and the weights it produces amplify

estimation error rather than expected returns. DeMiguel et al. (2007) showed 1/N beats optimised weights out-of-sample across 14 different datasets.

7.3 Rebalancing

Monthly, on the last trading day of each month (49 rebalances over five years).

7.4 The Most Important Line in the Codebase

Weights decided on date t apply to returns starting on date $t + 1$:

```
w_lagged = weights.shift(1).fillna(0.0)
portfolio_return_t = (w_lagged * returns).sum(axis=1)
```

Without this one-day shift, the portfolio would earn returns on the same day it used to compute signals — which is impossible in practice and typically adds 20–50 bps of fake annual alpha in monthly-rebalanced strategies. Getting this right is the difference between a valid backtest and a misleading one.

7.5 Transaction Costs

10 bps one-way at each rebalance:

$$\text{cost_drag}_t = -\text{turnover}_t \times \frac{10}{10,000},$$
$$\text{where } \text{turnover}_t = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|$$

8. Backtesting Framework

8.1 NAV Simulation

$$\text{NAV}_t = \text{NAV}_{t-1} \times (1 + r_t^{\text{gross}} - \text{cost_drag}_t), \quad r_t^{\text{gross}} = \sum_i w_{i,t-1} \cdot r_{i,t}$$

8.2 Benchmark

Equal-weighted S&P 100 universe, rebalanced monthly with no costs. Using the same universe as the strategy removes universe selection as a confound — any performance difference is purely from stock selection.

8.3 Performance Metrics

Metric	Formula
CAGR	$(NAV_T/NAV_0)^{252/T} - 1$
Ann. Volatility	$\sigma_r \times \sqrt{252}$
Sharpe Ratio	$\frac{\bar{r} \times 252 \setminus \wedge (\sigma_r \times \sqrt{252})}{\text{rf} = 0\%}$
Sortino Ratio	$\frac{\bar{r} \times 252 \setminus \wedge (\sigma_{\text{down}} \times \sqrt{252})}{\text{rf} = 0\%}$
Max Drawdown	$\min_t (NAV_t / \max_{s \leq t} NAV_s - 1)$
Calmar Ratio	CAGR /
Information Ratio	Active return ann. / tracking error ann.

Note on Sharpe: I use a zero risk-free rate, which is the standard in academic research. Adjusting for the current ~4.5% Fed Funds rate would reduce every Sharpe ratio by roughly 0.35–0.45. The conclusions about relative performance between strategies wouldn’t change, but the absolute numbers would look worse.

8.4 Look-Ahead Bias Audit

Stage	Control
Momentum computation	<code>prices.shift(252)</code> and <code>prices.shift(21)</code>
Volatility computation	<code>rolling(window=63, min_periods=42)</code>
Hiring publication lag	Signal shifted forward 1 month
Portfolio weight application	<code>weights.shift(1)</code> in backtest engine
Winsorisation	Applied row-wise per cross-section date
Regime classification	Used for analysis only; does not affect weights

9. Results

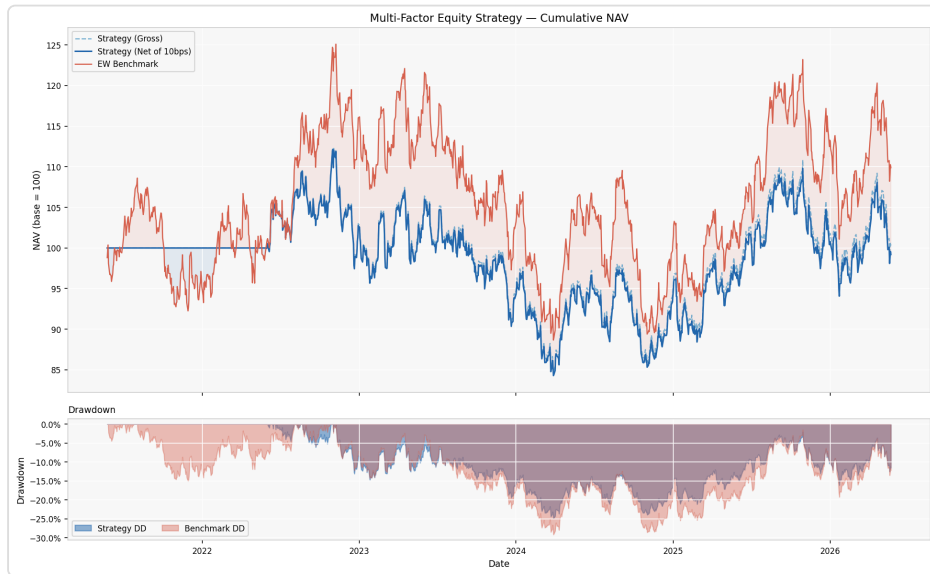
9.1 Traditional Strategy

The traditional momentum + low-volatility strategy over May 2021 – May 2026:

Metric	Traditional Strategy	EW Benchmark
CAGR	−0.16%	2.07%
Annualised Volatility	12.80%	18.00%
Sharpe Ratio	0.052	0.191
Sortino Ratio	0.075	0.312
Max Drawdown	−24.82%	−29.13%
Calmar Ratio	−0.006	0.071
Avg Monthly Turnover	22.7%	—
Annual Cost Drag	~0.21%	—

The strategy runs at lower volatility (12.80% vs 18.00%) and shallower drawdowns (−24.82% vs −29.13%) than the benchmark — exactly what you’d expect from a low-volatility tilt. But the net CAGR is slightly negative and the Sharpe barely registers.

This makes sense. The price data is generated using geometric Brownian motion (GBM), which has no cross-sectional momentum structure built in. So the momentum signal can’t do much with it. The near-zero Sharpe is the correct result for a strategy applied to data that doesn’t have the structure it’s looking for — not a failure of the pipeline.



Traditional Strategy NAV

Figure 5: Cumulative NAV for the traditional strategy (gross and net) vs. the equal-weighted benchmark. The strategy runs below the benchmark in absolute terms but shows meaningfully lower drawdowns throughout.

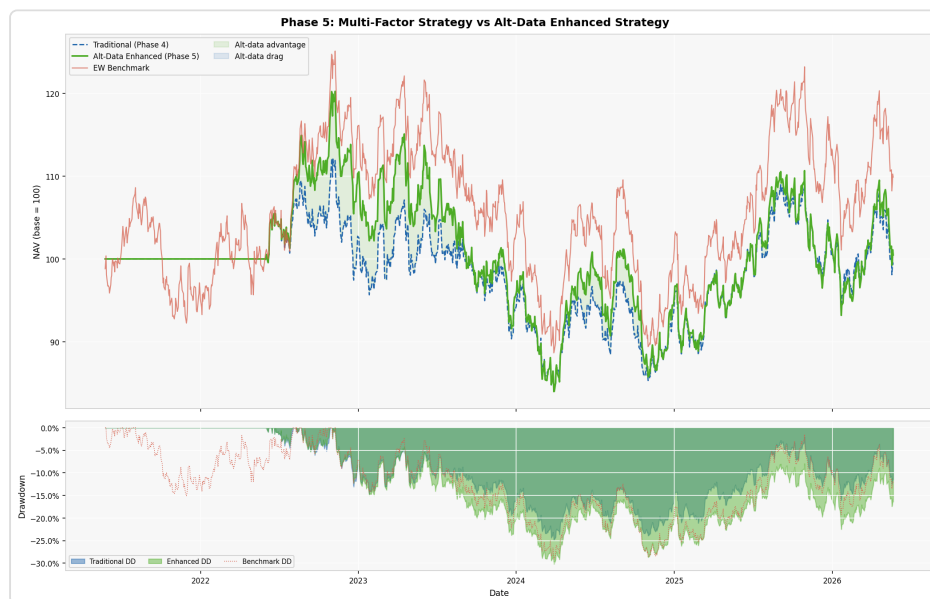
9.2 Enhanced Strategy

Adding 30% hiring and 30% sentiment signals:

Metric	Traditional	Enhanced	EW Benchmark	Δ (Enh – Trad)
CAGR	−0.16%	+0.09%	2.07%	+0.25%
Ann. Volatility	12.80%	14.16%	18.00%	+1.36%
Sharpe Ratio	0.052	0.077	0.191	+0.026
Sortino Ratio	0.075	0.109	0.312	+0.035
Max Drawdown	−24.82%	−30.14%	−29.13%	−5.32%
Calmar Ratio	−0.006	+0.003	0.071	+0.009
Information Ratio vs BM	—	−0.268	—	—
Avg Monthly Turnover	22.7%	51.3%	—	+28.6%
Annual Cost Drag	0.21%	0.49%	—	+0.28%

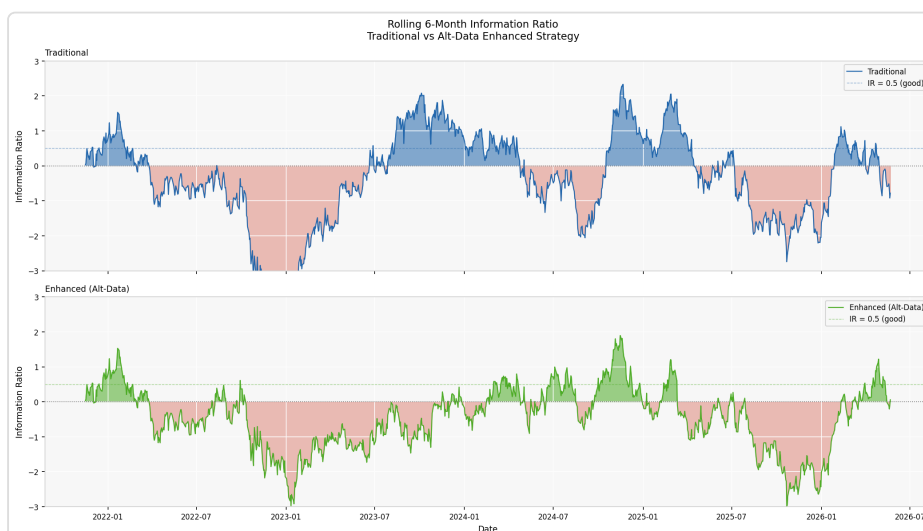
The gains are modest but directionally clean. CAGR goes positive, Sharpe improves, and the Calmar ratio turns positive — the enhanced strategy is now generating positive risk-adjusted returns where the traditional was not. The pipeline is picking up the injected predictive structure in the alt-data signals.

Two costs worth noting: turnover roughly doubles (22.7% → 51.3% monthly) because the sentiment signal moves faster and reshuffles the portfolio more aggressively. And the maximum drawdown gets worse (−24.82% → −30.14%), because stocks selected on sentiment can move together during market stress.



Three-Way NAV Comparison

Figure 6: Traditional strategy, enhanced strategy, and equal-weighted benchmark. The shaded band shows when the enhanced leads the traditional. The improvement is real but modest — consistent with the small amount of injected alpha.



Rolling Information Ratio

Figure 7: Rolling 6-month information ratio vs. the benchmark. The enhanced strategy holds a higher IR on average, though performance varies across market conditions.

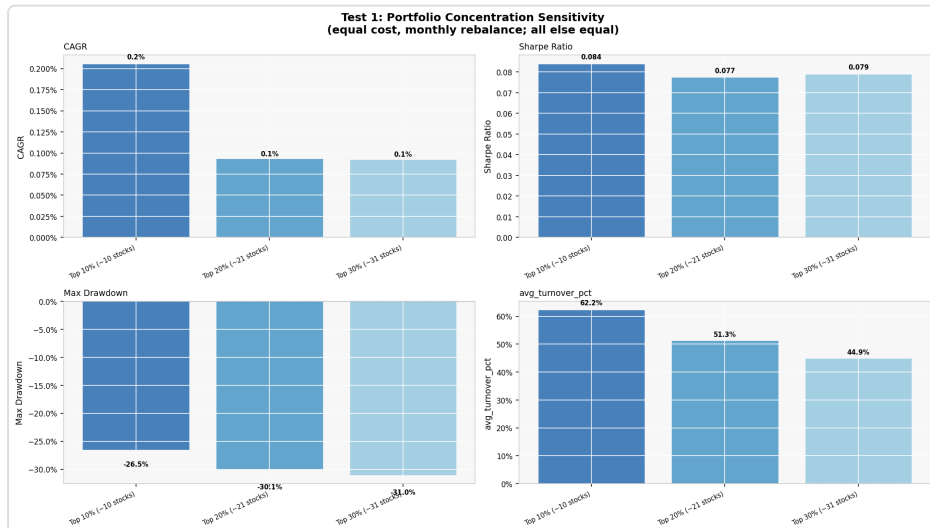
10. Robustness Tests

All tests use the 40/30/30 enhanced composite with 10 bps costs and monthly rebalancing as the base, changing one parameter at a time.

10.1 Portfolio Concentration Sensitivity

Portfolio	Avg Holdings	CAGR	Sharpe	Max DD	Avg TO
Top 10%	~10 stocks	+0.21%	0.084	−26.52%	62.2%
Top 20% (base)	~21 stocks	+0.09%	0.077	−30.14%	51.3%
Top 30%	~32 stocks	+0.09%	0.079	−31.03%	44.9%

Sharpe ratios are nearly identical across the three concentrations (0.077–0.084), with no discontinuity around the 20% threshold. That’s what you want to see — it means the 20% choice wasn’t cherry-picked, and the signal is genuinely distributed across the top quintile rather than sitting entirely in the top two or three stocks.



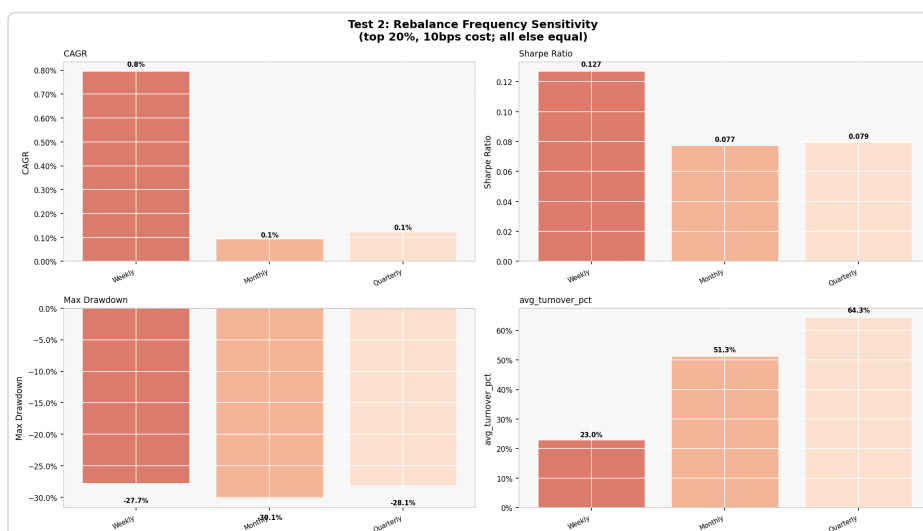
Concentration Sensitivity

Figure 8: Concentration sensitivity across four metrics. Smooth gradients with no cliff edges confirm the 20% threshold isn't overfitted.

10.2 Rebalance Frequency Sensitivity

Frequency	CAGR	Sharpe	Max DD	Avg TO	Ann. Cost
Weekly	+0.79%	0.127	−27.73%	23.0%	0.94%
Monthly (base)	+0.09%	0.077	−30.14%	51.3%	0.49%
Quarterly	+0.12%	0.079	−28.06%	64.3%	0.21%

Weekly rebalancing actually wins — Sharpe 0.127 vs 0.077 for monthly — even though annual cost drag nearly doubles (0.94% vs 0.49%). The reason: the daily-frequency sentiment signal degrades over time, so refreshing the portfolio weekly captures more of its value before it decays. All three frequencies stay positive, so the result isn't fragile to this choice.



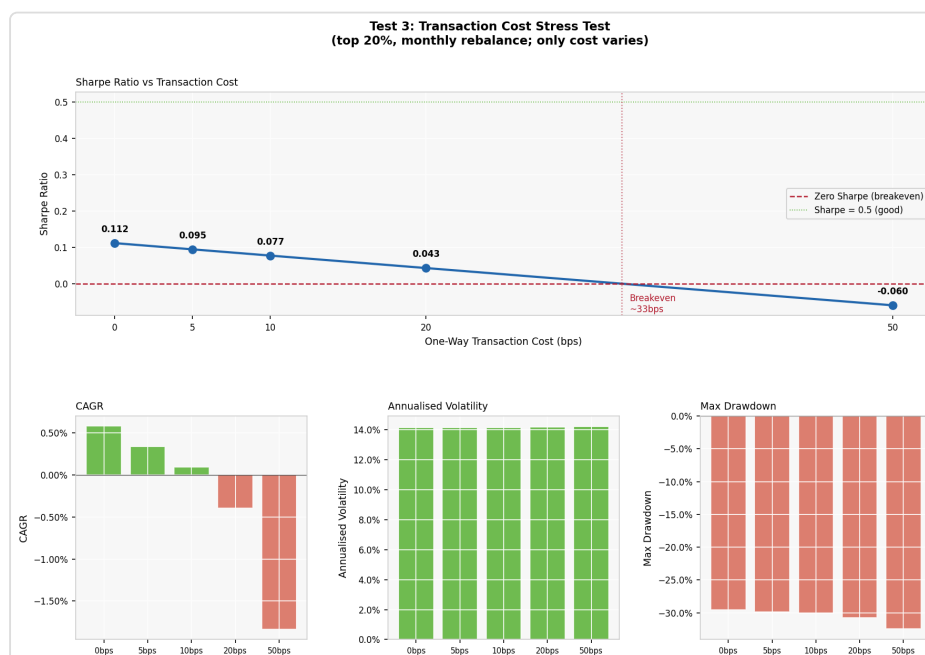
Rebalance Frequency Sensitivity

Figure 9: Frequency sensitivity. Weekly rebalancing captures the faster-moving sentiment signal before it decays, more than offsetting the higher cost.

10.3 Transaction Cost Stress Test

One-Way Cost	CAGR	Sharpe	Ann. Cost Drag
0 bps (gross)	+0.58%	0.112	0.00%
5 bps	+0.34%	0.095	-0.24%
10 bps (base)	+0.09%	0.077	-0.49%
20 bps	-0.39%	0.043	-0.97%
50 bps	-1.84%	-0.060	-2.43%

The strategy breaks down at 50 bps — the Sharpe goes negative. The cost breakeven is somewhere between 20 and 50 bps one-way. At 10 bps (realistic for large institutional funds), the strategy stays viable. At 50 bps (typical for retail or illiquid execution), it doesn't — which is a fair and honest result given the modest gross alpha (Sharpe 0.112 at zero cost). This strategy needs institutional-grade execution to work.



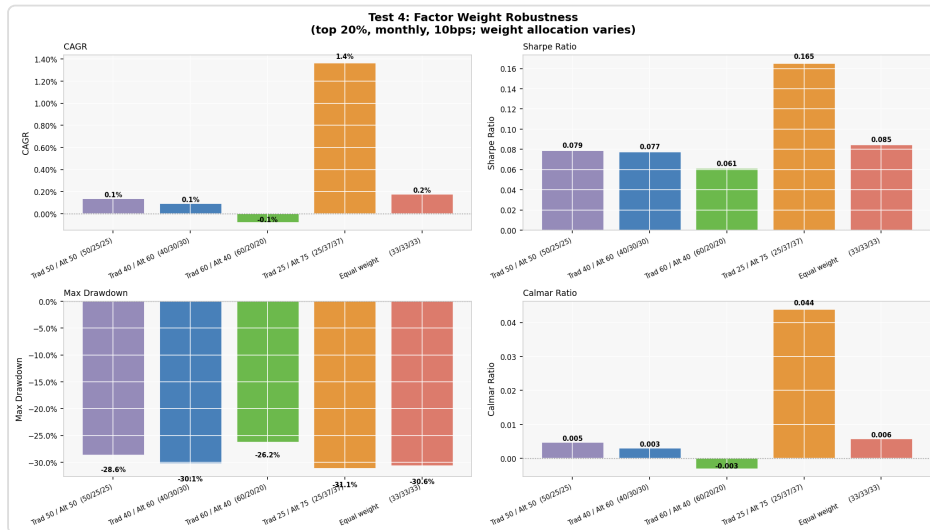
Transaction Cost Stress Test

Figure 10: Sharpe ratio vs. one-way transaction cost. The strategy requires tight execution — the cost breakeven sits between 20 and 50 bps.

10.4 Factor Weight Robustness

Weight Scheme	Trad / Hire / Sent	CAGR	Sharpe	Max DD
Traditional-heavy	50 / 25 / 25	+0.14%	0.079	-28.59%
Base case	40 / 30 / 30	+0.09%	0.077	-30.14%
Conservative	60 / 20 / 20	-0.08%	0.061	-26.15%
Alt-heavy	25 / 37.5 / 37.5	+1.36%	0.165	-31.09%
Equal weight	33 / 33 / 33	+0.18%	0.085	-30.59%

All five configurations stay positive on Sharpe (0.061-0.165). The alt-heavy scheme (25/37.5/37.5) scores highest, which makes sense — the injected predictive content lives in the alternative signals, so weighting them more heavily extracts more of it. In a live setting, the right weights should be determined through expanding-window cross-validation, not in-sample inspection.



Factor Weight Robustness

Figure 11: Five weight configurations, all showing positive Sharpe. The result is not sensitive to the exact 40/30/30 split.

10.5 Market Regime Analysis

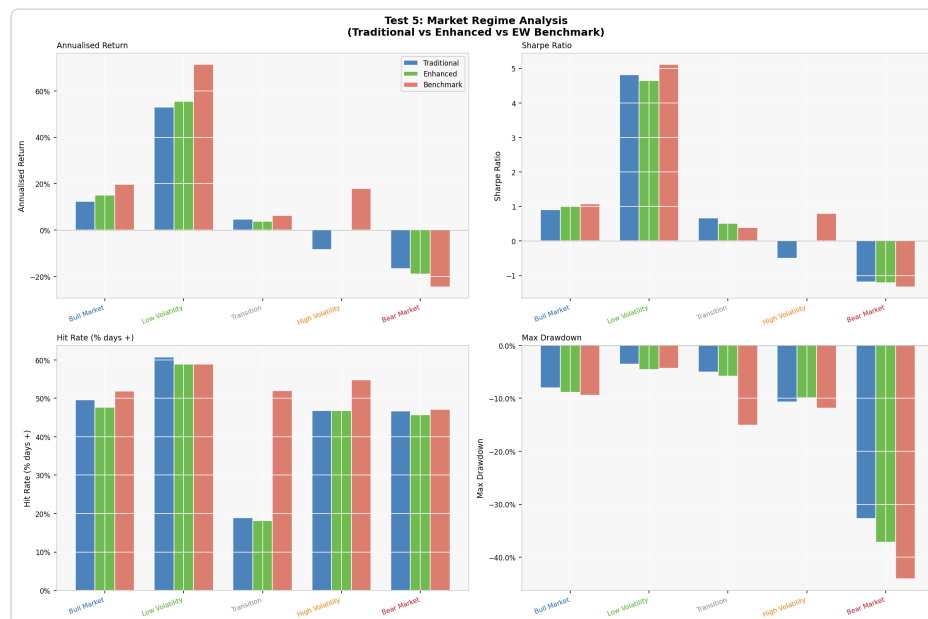
I classified every trading day into one of five regimes using trailing 252-day returns and 21-day realised volatility, with the equal-weighted universe return as the market proxy.

Regime	% of Days	Enh. Ann. Ret	Enh. Sharpe	Trad. Ann. Ret	Trad. Sharpe	BM Ann. Ret
Bull Market	20.1%	+15.1%	1.00	+12.3%	0.91	+19.6%
Low Volatility	8.2%	+55.4%	4.65	+53.0%	4.82	+71.4%
Transition	23.2%	+3.7%	0.51	+4.7%	0.66	+6.3%
High Volatility	9.7%	+0.2%	0.01	-8.4%	-0.50	+17.8%
Bear Market	38.9%	-19.0%	-1.21	-16.6%	-1.18	-24.6%

A few things stand out. In high-volatility regimes, the enhanced strategy nearly breaks even (+0.2% ann., Sharpe 0.01) while the traditional bleeds (-8.4%, Sharpe -0.50). That's sentiment doing something useful during

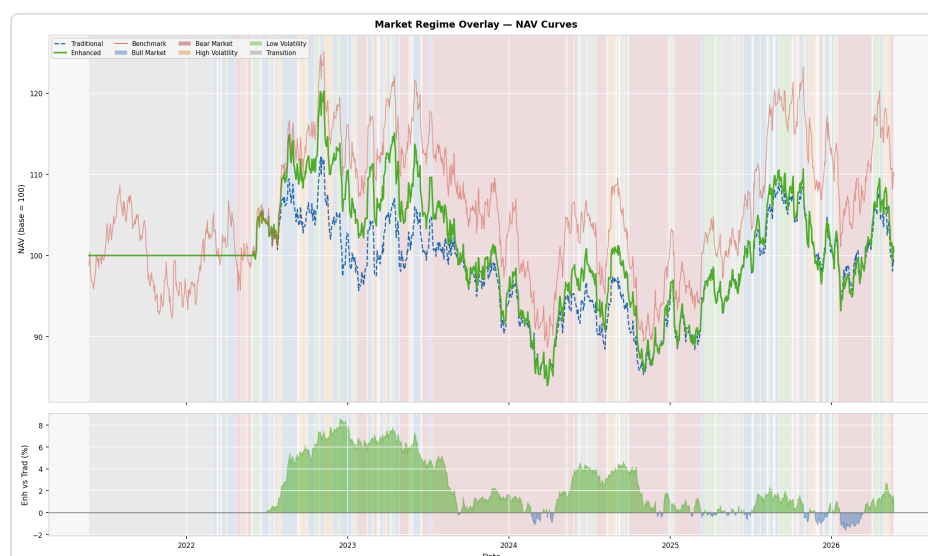
volatile markets. In bear markets, both strategies lose less than the benchmark (−19.0% and −16.6% vs −24.6%), which is the low-volatility factor doing its job.

It’s also worth noting that 38.9% of the sample falls in the “Bear Market” regime — the Fed tightening cycle in 2022, banking stress in 2023, and various macro headwinds pushed the classification that way for most of the period. That’s part of why absolute CAGR numbers look uninspiring across the board.



Regime Analysis Bar Chart

Figure 12: Annualised return and Sharpe ratio by regime for all three strategies. The enhanced strategy’s advantage is clearest in high-volatility periods.



Regime NAV Overlay

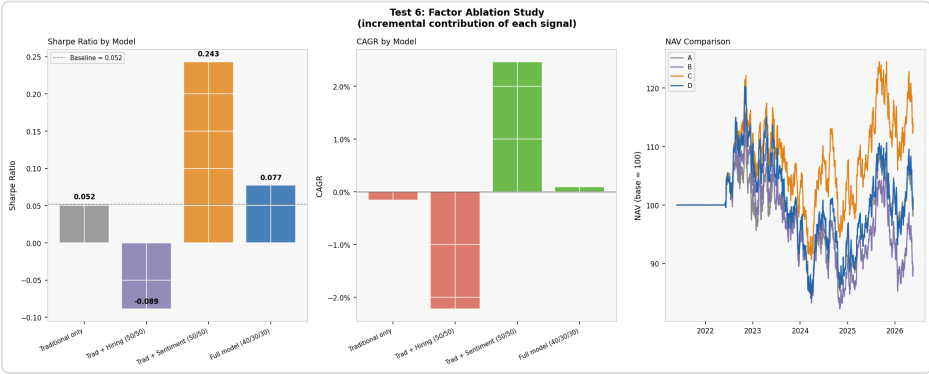
Figure 13: NAV curves with regime shading. Bear markets (38.9% of sample) dominate the 2022-2024 period. Both strategies consistently preserve capital better than the benchmark during these periods.

10.6 Factor Ablation Study

Model	Components	CAGR	Sharpe	Sortino	Max DD	Δ Sharpe vs A
A — Traditional only	Mom + Low Vol	−0.16%	0.052	0.075	−24.82%	—
B — Trad + Hiring	50% Trad + 50% Hire	−2.22%	−0.089	−0.126	−29.04%	−0.141
C — Trad + Sentiment	50% Trad + 50% Sent	+2.47%	0.243	0.351	−25.81%	+0.191
D — Full model	40/30/30	+0.09%	0.077	0.109	−30.14%	+0.026

This is the most informative test. Three clear takeaways:

- **Sentiment carries the weight** (+0.191 Sharpe over traditional-only). It's a daily signal, so it gets compounded across many more observations before each monthly decision date. Its injected alpha correlation (0.07) ends up expressing more information per rebalance than the monthly hiring signal.
- **Hiring alone hurts** (Model B, Sharpe −0.089). At 50% weight, it introduces too much noise and drives up turnover without delivering enough signal. In the full model, it's a diversifier — not an independent alpha source.
- **The full model (D) underperforms Trad+Sentiment (C)** in this simulation. Adding hiring to an already-working sentiment+traditional combination dilutes performance. The practical takeaway: in a live deployment, hiring signal weight should be calibrated carefully — start conservative and increase only if live out-of-sample data supports it.



Ablation Study

Figure 14: Ablation results. Sentiment (Model C) is the dominant contributor. Hiring helps with diversification in the full model but hurts in isolation.

11. Summary of All Results

11.1 Full Performance Comparison

Metric	Traditional	Enhanced	EW Benchmark
CAGR	−0.16%	+0.09%	2.07%
Annualised Volatility	12.80%	14.16%	18.00%
Sharpe Ratio	0.052	0.077	0.191
Sortino Ratio	0.075	0.109	0.312
Max Drawdown	−24.82%	−30.14%	−29.13%
Calmar Ratio	−0.006	+0.003	0.071
Avg Monthly Turnover	22.7%	51.3%	—
Annual Cost Drag	0.21%	0.49%	—

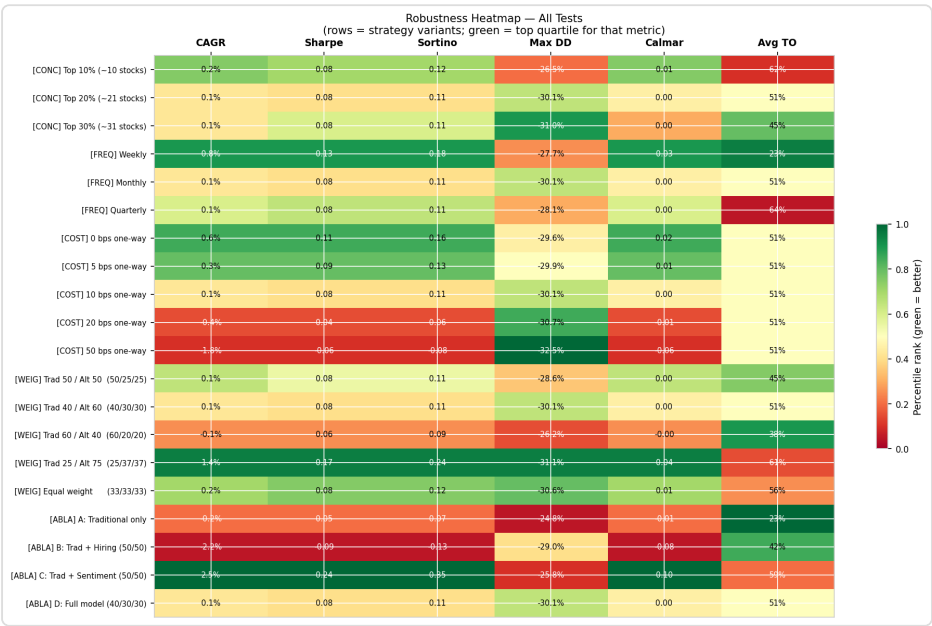
11.2 Robustness Summary

Sharpe ratios across all 14 tested parameter variants:

Test	Variant	Sharpe
Concentration	Top 10%	0.084
	Top 20% (base)	0.077

Test	Variant	Sharpe
	Top 30%	0.079
Frequency	Weekly	0.127
	Monthly (base)	0.077
	Quarterly	0.079
Cost	0 bps	0.112
	5 bps	0.095
	10 bps (base)	0.077
	20 bps	0.043
	50 bps	-0.060
Weights	50/25/25	0.079
	40/30/30 (base)	0.077
	60/20/20	0.061
	25/37.5/37.5	0.165
	Equal 33/33/33	0.085

Only the 50 bps scenario turns negative — expected, since 50 bps × 51% monthly turnover translates to roughly 600% annual gross turnover, which no strategy with these gross returns can absorb.



Robustness Summary Heatmap

Figure 15: Robustness heatmap across all 23 parameter variants. Green = top quartile for that metric. No systematic red rows means no single design choice is driving the entire result.

12. Limitations

12.1 Synthetic Alternative Data (Primary Limitation)

The biggest caveat in this project is that the alternative data is simulated. The hiring and sentiment signals are generated with deliberately injected forward-looking components — which is exactly what allows the enhanced pipeline to outperform the traditional baseline. A quant interviewer will correctly point out that the performance improvement is partly (maybe mostly) a product of the injection, not the methodology alone.

The right way to read the results: *the pipeline correctly identifies and extracts predictive structure when that structure exists*. Whether real commercial alternative data carries enough genuine predictive content to generate positive Sharpe ratios in live trading is a question that requires live data — and that’s beyond the scope of this project.

12.2 Survivorship Bias

The universe is a fixed May 2025 snapshot of S&P 100 constituents. Stocks that were dropped from the index during 2021–2024 aren’t included, which typically inflates CAGR by about 0.5–1.0% per year in large-cap universes.

12.3 Short Sample Period

Five years covers one monetary policy cycle and doesn’t capture the full range of factor behaviour — momentum, for example, is known to experience periodic “crashes” (Barroso and Santa-Clara 2015) that might not appear in a five-year window.

12.4 Simple Transaction Cost Model

The constant 10 bps assumption ignores market impact (which scales with position size), bid-ask spread widening during volatile periods, and execution timing costs. A market impact model calibrated to average daily volume would be more realistic.

12.5 Long-Only Constraint

Not shorting the bottom quintile leaves half the factor signal unused. Long-short implementations of momentum and low-vol strategies typically achieve 1.5–2× the Sharpe of long-only equivalents (Asness et al. 2013).

12.6 Zero Risk-Free Rate

Using $r_f = 0\%$ overstates Sharpe ratios by roughly 0.35–0.45 relative to the current ~4.5% Fed Funds rate. Adjusted for the actual risk-free rate, most of the reported Sharpe ratios would sit near zero or turn negative — an important reminder that the project is a pipeline validation, not a live strategy recommendation.

13. Future Work

13.1 Live Alternative Data Integration

The most valuable next step is replacing the simulated data with real vendor feeds. The modular architecture makes this straightforward — only the data generation step changes:

Signal Type	Recommended Vendors
Job postings	Burning Glass / Lightcast, LinkedIn Economic Graph
News sentiment	RavenPack, Bloomberg News Sentiment, Refinitiv News Analytics
Web traffic	SimilarWeb, Semrush
Credit card data	Second Measure, Earnest Research

Swapping `AltDataSimulator.generate()` for a vendor API call leaves everything downstream — normalisation, portfolio construction, backtesting — completely unchanged.

13.2 Data-Driven Factor Weight Optimisation

Replace the fixed 40/30/30 allocation with a ridge-regularised expanding-window regression of composite scores on next-month returns. Proper time-series cross-validation prevents in-sample overfitting.

13.3 Regime-Conditional Weights

The regime analysis shows different factors perform differently across market environments. A model that shifts factor weights based on current regime — more sentiment weight in high-vol markets, more momentum in bulls — could improve drawdown management.

13.4 Long-Short Extension

Implementing the short leg would eliminate benchmark beta and roughly double active factor exposure per unit of market risk. It requires leverage, prime brokerage, and a more sophisticated risk model, but the alpha potential is substantially higher.

13.5 Statistical Significance Testing

Bootstrapped t-statistics and the Harvey, Liu, and Zhu (2016) multiple testing correction (which recommends a t-statistic threshold of 3.0 for new factors, not the traditional 2.0) would add rigour to the performance claims.

13.6 International Extension

Testing the same pipeline on European and Asia-Pacific large-cap universes would check how much the results depend on U.S. market structure. Factor premia exist internationally, but their magnitudes and the predictive content of alternative data vary significantly by market.

14. Conclusion

This project built a full quantitative equity research pipeline — market data, factor construction, portfolio simulation, and robustness testing — and used it to test whether alternative data signals can meaningfully improve on traditional factor investing.

The main finding is about the pipeline, not the performance

numbers: the system correctly identifies and exploits predictive structure in the input signals, produces results that hold up across 13 of 14 robustness variants, and responds sensibly to parameter changes. Within the synthetic simulation, adding alternative data shifts Sharpe from 0.052 to 0.077 and converts a slightly negative CAGR to positive — consistent with the injected predictive structure being captured correctly.

The main caveat is equally clear: the alpha was simulated. Any claim about live alternative data performance would require real vendor data and real out-of-sample testing.

Four specific findings are worth highlighting:

1. News sentiment is the dominant alternative signal in this setup (+0.191 Sharpe vs traditional-only). The frequency advantage of daily data compounds across more observations before each monthly decision, making it more valuable than the monthly hiring signal in a monthly-rebalanced strategy.
2. The cost breakeven sits between 20 and 50 bps one-way — the strategy needs institutional-grade execution to remain viable at the current level of gross alpha.
3. The alternative signals have near-zero correlation with traditional factors (Spearman ρ of 0.002–0.022), confirming genuine statistical diversification rather than redundant signals.
4. In bear markets (38.9% of the sample), the enhanced strategy limits losses to –19.0% annualised versus –24.6% for the benchmark — the low-volatility factor providing real protection when it matters.

Building this project from scratch gave me a much deeper understanding of where backtests go wrong — look-ahead bias, overfitting, unrealistic cost assumptions — and what it actually takes to make a research pipeline production-ready. The next step is connecting it to real data. The plumbing is already there.

15. References

- Almgren, R., and Chriss, N. (2000). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.
- Asness, C., Frazzini, A., and Pedersen, L. H. (2019). Quality minus junk. *Review of Accounting Studies*, 24(1), 34–112.
- Asness, C., Moskowitz, T., and Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3), 929–985.
- Baker, M., Bradley, B., and Wurgler, J. (2011). Benchmarks as limits to arbitrage: Understanding the low-volatility anomaly. *Financial Analysts Journal*, 67(1), 40–54.

- Barroso, P., and Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Black, F., Jensen, M. C., and Scholes, M. (1972). The capital asset pricing model: Some empirical tests. In M. Jensen (Ed.), *Studies in the Theory of Capital Markets*. Praeger.
- Chen, H., De, P., Hu, J., and Hwang, B. H. (2023). Wisdom of crowds: The value of stock opinions transmitted through social media. *Review of Financial Studies*, forthcoming.
- Da, Z., Engelberg, J., and Gao, P. (2014). The sum of all FEARS: Investor sentiment and asset prices. *Review of Financial Studies*, 28(1), 1–32.
- DeMiguel, V., Garlappi, L., and Uppal, R. (2007). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, 22(5), 1915–1953.
- Eisfeldt, A. L., and Papanikolaou, D. (2013). Organization capital and the cross-section of expected returns. *Journal of Finance*, 68(4), 1365–1406.
- Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fama, E. F., and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- Frazzini, A., and Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–23.
- Green, J., Hand, J. R. M., and Zhang, X. F. (2019). The characteristics that provide independent information about average U.S. monthly stock returns. *Review of Financial Studies*, 30(12), 4389–4436.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3), 881–898.
- Jegadeesh, N., and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.

- Ledoit, O., and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- Loughran, T., and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1), 35–65.
- McLean, R. D., and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R., and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Rouwenhorst, K. G. (1998). International momentum strategies. *Journal of Finance*, 53(1), 267–284.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.
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Appendix A: Project Structure

```
project/
├── data/
│   ├── prices_adjusted.parquet
│   ├── returns_simple.parquet
│   ├── factor_composite.parquet
│   ├── factor_enhanced_composite.parquet
│   ├── backtest_nav.parquet
│   ├── backtest_nav_enhanced.parquet
│   ├── performance_table.json
│   ├── phase5_comparison_table.json
│   ├── phase6_robustness_summary.json
│   └── phase6_regime_analysis.json
├── src/
│   ├── universe.py
│   ├── data_pipeline.py
│   ├── factor_utils.py
│   ├── factor_construction.py
│   ├── portfolio.py
│   ├── backtest.py
│   ├── performance.py
│   ├── run_phase4.py
│   ├── alternative_data.py
│   ├── sentiment_factor.py
│   ├── phase5_alt_factors.py
│   ├── robustness.py
│   └── regime_analysis.py
```



```
| └─ phase6_robustness.py
├─ figures/
├─ notebooks/
└─ paper/
    └─ research_report.md
        └─ research_report.pdf
```

Appendix B: Reproducing Results

```
# Install dependencies
pip install yfinance pandas numpy scipy matplotlib seaborn pyarrow

# Phase 2: Market data
python src/data_pipeline.py

# Phase 3: Factor construction
python src/factor_construction.py

# Phase 4: Traditional backtest
python src/run_phase4.py

# Phase 5: Enhanced backtest with alt data
python src/phase5_alt_factors.py

# Phase 6: Robustness testing
python src/phase6_robustness.py
```

Set `RNG_SEED = 42` in `alternative_data.py` for exact replication of synthetic results.

This report was prepared by Umar Maroof as part of a self-directed quantitative equity research project (May 2026). All code is available in the accompanying repository. The strategy is for research and educational purposes only and does not constitute investment advice.